Practice Exercise 3

Problem: You have a dataset containing information about customers of a retail store, including their age, gender, annual income, and

spending score. Your task is to cluster customers based on their similarities and identify distinct customer segments.

Steps:

1. Data Exploration.- Analyze the dataset for missing values, outliers, and distributions of features.- Visualize the data using scatter plots or histograms to identify patterns.

2. Feature Scaling. Normalize or standardize the features (age, annual income, and spending score) to ensure they have a similar scale.

3. Clustering.- Choose a clustering algorithm (e.g., K-means, K-medoids, DBSCAN, Hierarchical clustering,...).- Experiment with different parameters and evaluate the results using appropriate metrics.

4. Evaluation.- Use metrics like silhouette coefficient to assess the quality of the clusters.- Visualize the clusters using scatter plots or dendrograms to understand the segmentation